BEACON-V ML-Assisted Delay-Tolerant Emergency Message Delivery in Vehicular Networks

This document explains the BEACON-V project in full detail, starting from basic concepts and gradually building up to the complete system. It is written for readers with no prior knowledge of vehicular networks, DTNs, or machine learning.

1. Background Concepts

1.1 What happens when the internet fails?

During disasters such as earthquakes, floods, or large-scale accidents, mobile towers and internet infrastructure often fail. However, people still carry smartphones or vehicles that can communicate locally using Bluetooth or Wi-Fi. BEACON-V uses this local communication to deliver emergency messages even without the internet.

1.2 What is a Delay-Tolerant Network (DTN)?

A Delay-Tolerant Network is a type of network designed to work when continuous connectivity cannot be assumed. Nodes store messages, carry them while moving, and forward them when they meet other nodes. This is known as the store-carry-forward paradigm.

1.3 What is a Vehicular DTN (VDTN)?

A Vehicular DTN is a DTN where vehicles (cars, buses, or phones inside vehicles) act as nodes. Vehicles naturally move around a city, making them excellent carriers for messages.

2. Problem Definition

The main challenge is deciding which vehicle should carry an emergency message. Traditional DTN routing protocols often copy messages randomly or excessively, wasting bandwidth and increasing delay. BEACON-V aims to forward messages only to vehicles that are both trustworthy and likely to reach the destination.

3. Core Idea of BEACON-V

BEACON-V improves a classical routing method called Spray-and-Wait. Instead of blindly spraying messages to all neighbors, BEACON-V uses two Machine Learning models to make intelligent decisions.

1. Movement Model: Predicts whether a vehicle is likely to reach the destination area.
2. Trust Model: Detects malicious or unreliable vehicles.

4. Data Used in the Project

The project uses real-world vehicle GPS data such as the Beijing T-Drive dataset. Each record contains vehicle ID, timestamp, latitude, and longitude. This data represents realistic urban driving behavior.

Because the dataset is extremely large (millions of rows), all processing is done using memory-safe chunked pipelines that process small portions at a time.

5. Data Preprocessing Pipeline

1. Sort GPS points by vehicle and time.
2. Remove duplicate and invalid records.
3. Compute speed, acceleration, and jerk.
4. Detect stops using speed thresholds.
5. Compute dwell time for stop events.
6. Add temporal features such as hour and weekday.
7. Add semantic POI features using TF-IDF.
8. Split data into train, validation, and test sets by time.

6. Prediction Scenarios

6.1 Scenario 1 – Stop Prediction

This task predicts whether a vehicle will stop within a short future window (e.g., 60 seconds). Stopping vehicles are poor message carriers because they stop moving.

6.2 Scenario 2 – Destination Zone Entry Prediction

This task predicts whether a vehicle will enter a target geographic zone such as a hospital. This directly determines whether a vehicle is a good carrier for emergency messages.

6.3 Scenario 3 – Trust Prediction

This task identifies malicious or faulty vehicles by analyzing abnormal behavior such as unrealistic speeds or message dropping.

7. Machine Learning Models

The Movement Model is implemented using a Long Short-Term Memory (LSTM) neural network. It processes sequences of recent GPS and motion features to predict future behavior. The Trust Model is a lightweight classifier designed for fast inference on mobile devices.

8. Routing Decision Flow

1. Vehicles discover neighbors using Bluetooth beacons.
2. Vehicles exchange recent movement and trust features.
3. Movement score is computed using the LSTM model.
4. Trust score is computed using the trust classifier.
5. Messages are forwarded only if both scores exceed thresholds.
6. Messages are delivered when a vehicle enters the destination zone.

9. Evaluation and Results

BEACON-V is evaluated using simulations and real-device demos. It is compared against classical DTN routing protocols such as Epidemic Routing and basic Spray-and-Wait. Results show improved delivery rates and lower delays, especially in sparse networks.

10. Final Summary

BEACON-V demonstrates that combining realistic mobility modeling, trust assessment, and intelligent routing decisions can significantly improve emergency communication in disconnected vehicular environments.